MOOTER · SYSTEM DESIGN V2 · 2026-07-06

Your LLM router.
Local-first. Learns forever.

The canonical ecosystem map: how a prompt becomes a routing decision in <50 ms at $0, why the cloud only gets called when it earns its cost, where every component actually lives — and where this is going: pillars, roadmap, risks.

T0 local Ollama · $0 | T1 Haiku | T2 Sonnet | T3 Opus · risk floors | T5 Fable · opt-in @fable only

Five principles: no proxy · zero LLM cost to classify · doctrine > configuration · explainability non-negotiable · optimisation never overrides the doctrine.

Frozen invariant: `tools/router/classify.js` — sha256 CI-enforced. Learning happens around it, never inside it.

0 · Executive summary — honest, dated numbers only

METRIC	VALUE	DATE / SOURCE
"Mooter audited Mooter" real session	**$2.04 vs $11.78 = 82.7% saved**	2026-06-06
Live dashboard	$25.95 saved / 658 calls (47%)	2026-06-08 · hub live
Statusline, real day	saved $2.51 today (84% vs all-Opus) · **70% of turns local**	2026-06
Classification	<50 ms · $0 · hook p50 113 ms / p95 407 ms	2026-05
Pastor Wave 1	recall 100% · p99 3.74 ms	2026-05-27
Workflow demo (25 agents)	**$0.0028 real** (160× under estimate)	2026-06-07 · v1.16.0
Realistic savings envelope	65–82% vs all-Opus (not blogs' 95%)	FLOWCHART.md §0
Adversarial benchmark	in-domain COMPETITIVE · OOD dominated · **risk-axis BEST**	2026-05-24 tri-axis

Positioning (deliberately narrow): not an API proxy (LiteLLM/OpenRouter own that). Mooter is the **cost control plane for the solo dev / vibe coder** — no direct competitor in the niche; the founder IS the target user; the doctrine IS the product.

Mission

"Your LLM router. Local-first. Learns forever." (2026-06-07)

The soul

give a non-dev creative the power of Claude Code without the operational burden — perfect handoffs, auditable memory, GPU as $0 workforce.

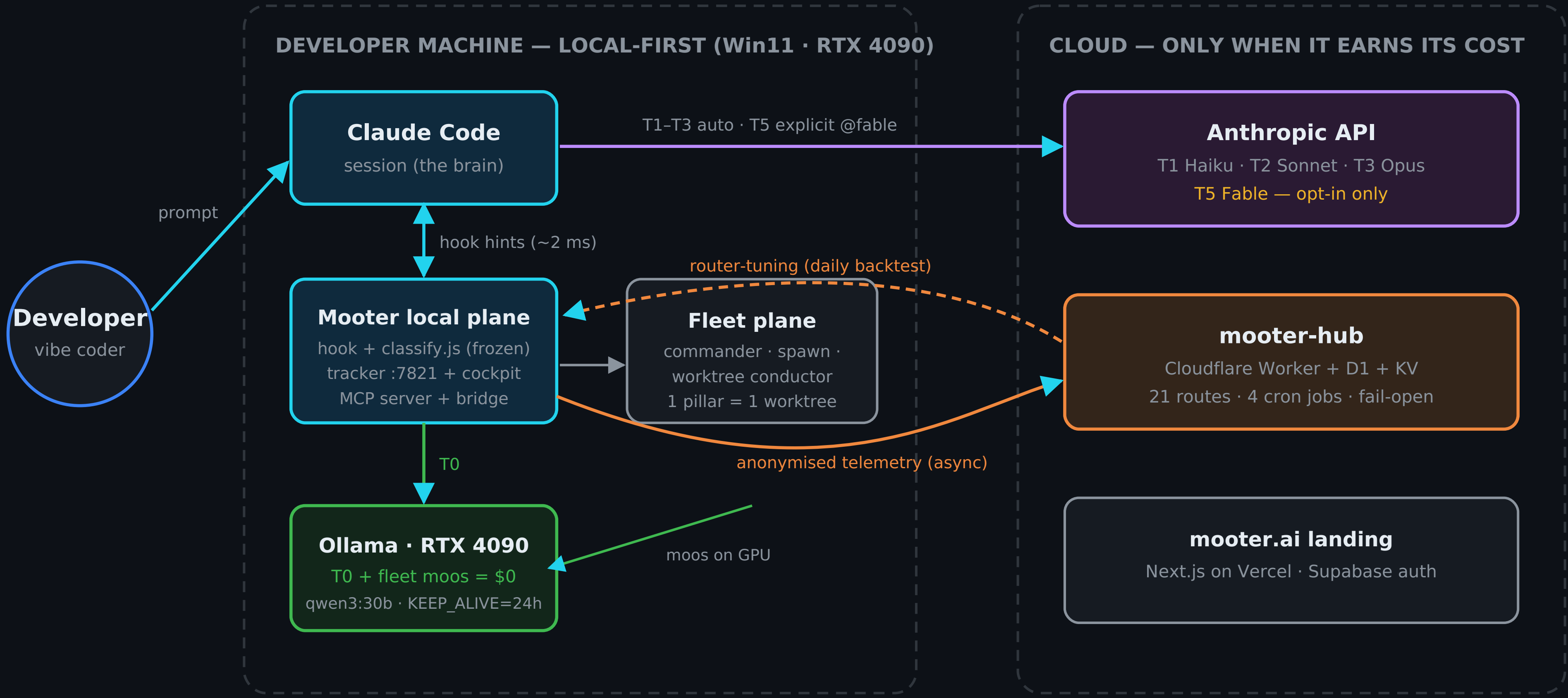
Moats

private dataset of real routing decisions · evolving tuning state · codified judgement · internal ecosystem as first customers · $0 self-improvement fleet on owned hardware.

Stage & biggest gap

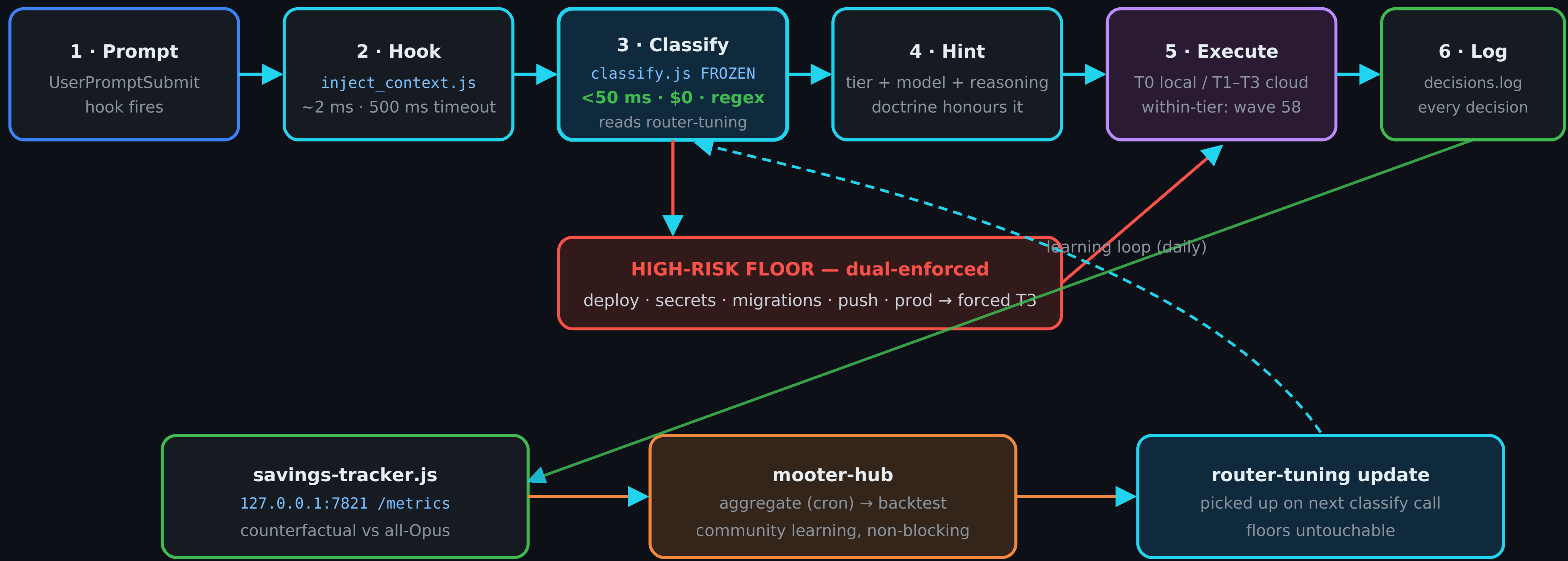
OSS (MIT), building in the open, 30+ waves shipped — never publicly launched. The gap is distribution execution, not technology (Risks, p. 10).

1 · System context — local-first, cloud when it earns its cost



Every arrow is fail-open: hub down, tracker down or Ollama cold never blocks a session. Mooter emits hints — it is never a proxy. The operational quartet: Cowork (design) · CC (irreversible) · moos ($0 workforce) · Mooter (conductor).

2 · Runtime flow — one prompt, end to end



Statusline proof: savings shown live — e.g. `saved $2.51 today (84% vs all-Opus)`, 70% of turns local in real sessions.

3 · Component map (real paths) & the L0–L16 layer stack

COMPONENT	PATH	ROLE
Classifier (frozen)	`tools/router/classify.js`	tier decision, <50 ms, regex, $0
Hook	`tools/router/inject_context.js`	spawns classifier, injects hint
Savings tracker	`tools/router/savings-tracker.js`	daemon :7821, counterfactual savings
Router engine + wave 58	`packages/router/src/`	within-tier selection, TES, adaptive matrix
CLI	`packages/cli/`	init · audit · wave · pastor
Cockpit v0.16.x	`packages/vscode-extension/`	lenses + Live Preview (dev server iframe)
Hub	`hub/ · wrangler.mooter.toml`	CF Worker + D1 + KV, 21 routes, 4 crons
MCP server	`packages/mcp-server/ (20 tools)`	Cowork/Desktop integration
Fleet plane	`packages/fleet-commander/ +4`	local moos on 4090, 1 pillar = 1 worktree
Pastor	`packages/router/src/adapter/`	TF-IDF priors + LoRA forge (Eixo 2/3)
Landing	`landing/`	Next.js · Vercel · Supabase
Bridge	`packages/mooter-bridge/`	Cowork ⇄ Mooter sessions (MCP stdio)

Layer stack (V4 + V5)

L	LAYER
`L0–L6`	cache · guardrails · features · kNN · confidence · dispatch · cascade
`L7–L11`	personalisation · fingerprint · skill graph · arbitrage · federated
`L12`	LLMLingua compression (opt-in)
`L13`	LoRA hot-swap (foundation)
`L14`	Setup Intelligence (Vector A)
`L15`	Ecosystem Awareness (Vector B)
`L16`	Prompt-Quality Intelligence (Vector C) — absorbed V4 telemetry "Layer X"

Wave 58 is orthogonal: specialization-matrix, decide-agent, tes-calculator, adaptive-learner, fable-5-routing operate **after** the tier is chosen — within-tier model/agent selection. New files only; no frozen engine file modified.

4 · Data stores & the three learning loops

STORE	LOCATION	WRITTEN BY → READ BY
Decision journal	`~/.claude/tools/router/decisions.log`	hook per turn → backtest, Pastor, cockpit
Router tuning	`~/.claude/tools/router/`	releases (/mooter-update + sync-hooks.js) → classify.js
Preferences	`~/.mooter/preferences.json`	user → statusline, cockpit
Cost-perf outcomes	`~/.mooter/cost-perf-log.jsonl`	executions → adaptive-learner
Adaptive matrix	`~/.mooter/specialization-overrides.json`	adaptive-learner (≥5 outcomes, EWMA) → decide-agent
Community aggregate	`hub D1 + KV`	events/aggregate → daily backtest
Fleet state	`_handoff/fleet/*/STATE.json + ledger + heartbeat`	orchestrator → conductor, cockpit, cronista

Privacy posture: telemetry anonymised + async; the session never depends on the cloud. Details: PRIVACY.md · SECURITY.md · ARCHITECTURE.md §14.

Three loops, deliberately separate

1 · Local, per-user
adaptive-learner EWMA over real outcomes; fills within-tier cells only; empty by default (honest).

2 · Community
hub aggregates deltas daily → new router-tuning → every install picks it up next classify. Fail-open.

3 · Pastor (specialisation)
deterministic TF-IDF priors + LoRA adapter forge trained on decision journals; served via hub pastor-v2.

Invariant across all three
learning may only move work down the ladder or sideways — high-risk floors and the frozen classifier are untouchable.

5 · Trade-offs — the interview-grade answers

DECISION	TRADE-OFF ACCEPTED	WHY IT WINS
Regex classifier, frozen	less "smart" than LLM/embedding routing	deterministic, <50 ms, $0, auditable — an LLM routing LLMs is recursive cost
Hints, not proxy	cannot force a model	zero added failure surface; Claude Code works even if Mooter dies
Local-first T0	TTFT ~4.6 s cold vs 0.7 s Haiku	$0 marginal cost; wins on long outputs; KEEP_ALIVE=24h kills cold start
Doctrine floors, dual-enforced	forgoes some savings	push/deploy/secrets are asymmetric-risk — cheap mistakes aren't cheap there
T5 opt-in only	no auto access to the strongest model	cost control + explicit human intent for the top tier
Hub async, fail-open	community learning lags up to a day	session latency is sacred; the cloud is an optimisation, not a dependency
Fleet capped ($5/day · quotas · human gate)	slower autonomous progress	autonomy without governance is how repos die; two-factor on destructive
Niche > proxy market	smaller initial TAM	LiteLLM/OpenRouter own the proxy game; nobody owns solo-dev cost control; founder is the user

Measured, not promised

65–82% realistic cost reduction vs all-Opus (not the 95% blogs promise) · workflow demo real cost $0.0028 (160× cheaper than estimated) · statusline live savings counterfactual.

Where depth lives

`ARCHITECTURE.md (core, source of truth)`
`docs/strategy/ARCHITECTURE_V4 + V5 (layers)`
`docs/strategy/MOOTER_ROADMAP.md (waves)`
`docs/adr/ (decisions) · INFRA.md (endpoints)`

Keeping it honest

Component changed → same-PR update of SYSTEM_DESIGN.md · new subsystem = 1 row + 1 arrow · never numbers you didn't measure.

6 · The self-improvement fleet — 12 + 1 pillars, $0 on owned GPU

Local moos run bounded rounds per pillar under fleet.json caps (GPU-heavy 1 · cloud 2 · pool 4 · $5/day · day quotas) with a human gate on anything irreversible. Charters = the roadmap wave each pillar serves.

PILLAR	SQUAD	CHARTER	MEASURED SUCCESS CRITERION
council · F3	Auto-Evolution	quality + calibration without regression	`oracle_gap ≤5% · p99 ≤100ms`
seguranca · F3	Security	continuous 3-promises + supply-chain audit	`0 leaks · audit 100%`
cronista · F3 (13th, meta)	Obs	scribe/QA: record all, cross-pillar harmony, pre-cook handoffs	`digest ≤1 round behind`
bench-eval	Obs	honest OOD eval + fix moo-verify	`ROUTER_SCORE/round`
matriz	Routing	specialization-matrix vs oracle + subagents→local (37% of burn)	`acc delta · tokens avoided`
quantizacao	Routing	AWQ/quant candidates vs qwen3:30b	`tok/s +X%, no quality loss`
integracoes-llm	Routing	draft→verify bench + MCP distiller (24% of burn)	`lossless speedup`
lora-dora	Auto-Evolution	1 real adapter, anti-forgetting	`passes forge gate`
vscode-plugin	Cockpit	additive polish + Context Guardian (66% of burn = ctx>150k)	`0 dead buttons`
design	Cockpit	hardcoded colours → var(--vscode-*)	`0 hex · 3 themes`
statusline	Cockpit	honest opt-in chips, default untouched	`default byte-identical`
site	Distribution	Install-Ready + educational onboarding	`Lighthouse ≥95 · install E2E`

Honesty note (2026-07-06): foundation is in main, git-proven — but all past rounds were DRY_RUN; FASE3 builds the real local runner.

7 · Where it is (2026-07-06) & where it goes

Current state

FRONT	STATE
Engine/router	OK prod, frozen, waves 26–34.5 shipped (v1.15→v1.21 tag line)
Value proof	OK measured — 82.7% audit · 47% dashboard · 70% local
Cockpit 0.16.x	~ installed locally; Marketplace publish = W3, pending
Live Preview	OK MP2+MP4 prod · MP5 select-to-edit in flight (NOW)
Fleet	next FASE3 ready (council+seguranca+cronista); 0 real rounds yet
Security	~ tracker guard v0.8.0 + pack auditor staged, pending merge
Distribution	NO never launched — the P0 of P0s

Bets in flight (specs ready in _handoff/)

Live Edit MP5 (select-to-edit router-native) · CTO Command Deck W15 (exception inbox + 4 lenses) · Mooter Mirror (Cowork brain in cockpit) · Conductor (session fleet dispatch) · **Moove** (Lovable→CC+Mooter migration as GTM funnel) · Overclock Fase 2 · Quota-Aware Routing (MP-Q, first in exec order).

Roadmap v3 — 8 squads (MOOTER_ROADMAP.md)

PHASE	WAVES	THEME
NOW P0	`W10K W13~ W15 W14~ W2 W3`	land polish · Delivery Cockpit · CTO Deck · comms · housekeeping · **distribution**
NEXT P1	`W5 W4 W9 W7 W6 W16`	Loop Sessions · Evolution Fleet · TTL distiller · Adapter Forge · Budget Cockpit · Live Preview cinema
FRONTIER P2	`W11 W8 W10 W12`	bandit+AWQ · speculative edge-cloud · graph routing · differential privacy

Priorisation rules: performance/cost first · leverage · $0 local speed · proof > promise · **land > start** (WIP down is rule #1). Max speed = max delegation to local.

Backlog P0 highlights (Notion): First-Magic Onboarding <5 min · Marketplace publish · hero repositioning "router-that-saves → control plane of the fleet" · Moove funnel · Mission Control tab · GPU-always-saturated policy · QA-funnel.

8 · Risks & honest gaps — stress-test these

#	RISK	EVIDENCE	MITIGATION IN MOTION
R1	**Distribution debt** — 30+ waves shipped, never launched; gate 2026-05-26 missed by non-execution, not traction	vault status history	W3 = P0 · First-Magic Onboarding · Moove funnel
R2	**OOD weakness** — dominated out-of-domain as generalist router	tri-axis 2026-05-24	double down on niche (risk-axis BEST) · bench-eval pillar runs honest OOD
R3	**Bus factor = 1** — solo non-dev founder, AI-leveraged	—	radical doc discipline (this doc, ADRs, 4 canónicos) · fleet automates maintenance
R4	**Fleet is promise until rounds are real** — all history DRY_RUN	fleet-heartbeat.json	FASE3 builds real runner · success criteria measured, not vibes
R5	**Security hardening staged, not merged** (tracker CSRF guard, pack auditor)	SYNC.md §SEC-1/2	CC checklist ready · supply-chain wave planned
R6	**Windows/mount fragility** — unreliable mount git state; past NUL-corruption	SYNC.md 2026-06-23	reads cross-checked · git writes always native
R7	**Single-vendor cloud** (Anthropic T1–T5)	—	multi-provider designed (Gemini/Codex/OpenAI wrappers exist) · T0 vendor-free
R8	**Change ≠ improvement** — council shipped without quality eval	2026-06-22	council pillar charter IS that eval · Friday eval harness

Why believe the numbers

Every §0 metric has date + source: decisions.log, hub endpoints live, benchmark bundles (frozen HEAD ce08f72c), tagged releases. Counterfactual method: what would all-Opus have cost for the same turns.

Keeping this honest

Component changed → same-PR update · new pillar = 1 row + 1 arrow · new number only with date + source, old numbers superseded, never silently replaced.